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Understanding the impacts of Electric Vehicle (EV) adoption over Gasoline-based vehicles on National Carbon Emissions: A study in Bangladeshi Perspective

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This thesis report was submitted in defence in fulfillment of the
Requirements for the Bachelor of Science Degree
In Computer Science and Engineering

April 2026

Jashore University of Science and Technology

Jashore-7408, Bangladesh



This thesis titled "Understanding the impacts of Electric Vehicle (EV) adoption over Gasoline-based vehicles on National Carbon Emissions: A study in Bangladeshi Perspective" submitted by Md. Mahmudul Amin Shakil for defence for the degree of Bachelor of Science in Computer Science and Engineering on April 2026

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Dedication

This thesis is dedicated to my parents, who have been a continuous source of inspiration for me over my journey with their steadfast support and encouragement. Their belief in my abilities and constant inspiration made this work possible. Your confidence in me and your advice have been crucial in getting me to this point.

Acknowledgments

First and foremost, I would like to express my deepest gratitude to my thesis supervisor, Monishanker Halder Sir, without whose guidance and expertise this work would not have been possible. Their insightful advice, continuous encouragement, and patience throughout the research process were invaluable. I also extend my thanks to the members of my thesis committee and the faculty of my department for their constructive feedback and support, which greatly improved the quality of this research. I am grateful to my friends for stimulating discussions and assisting in various stages of the project. Their cooperation and moral support created a motivating environment to overcome challenges. Finally, I wish to thank my family for their unwavering love and encouragement. Their understanding and sacrifices provided me with the strength to persevere and complete this thesis.

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Abstract

With the rapid growth of population, transportation demand has significantly increased in the urban areas of developing countries like Bangladesh. Major cities are facing transport-related challenges like affordable public transport. Public transport is generally shared by the public. To meet this demand, both cost-efficient and environmentally friendly solutions are needed, and in this scenario, battery-powered vehicles like three-wheeler easy bikes, electric rickshaws, and electric vehicles became a part of the urban transportation system as an alternative to the traditional engine vehicles [1]. In recent times, the number of battery-powered vehicles has massively increased due to their accessibility and comfort [2]. But these battery-powered vehicles are limited by a short operational range (60–100 km per charge), leading to frequent downtime. This thesis proposes how battery-powered vehicles contribute to minimizing the carbon emissions emitted from gasoline-based vehicles and the impacts of electric vehicles (EVs) [3]. In this thesis, I will consider many aspects for the adoption of the people of our country. The approach involves developing a high-fidelity traffic simulation model of Jashore using SUMO (Simulation of Urban Mobility), incorporating real-world vehicle data, and validating a vehicle-to-infrastructure (V2I) data collection architecture that captures dynamic emission data. This dataset will be used to train four machine learning (ML) algorithms, like linear regression (LR), random forest (RF), gradient boosting (GB), and XGBoost, to predict how much carbon is emitted in the environment in Jashore city. This software-based strategy is expected to significantly know about carbon emissions from different types of fuel-based vehicles on Jashore roads, substantially benefiting the sustainability and efficiency of urban transport in Bangladesh. Among these models, Random Forest performed well with the highest accuracy (93.15%). This research emphasizes the entire life cycle emissions of fuel-based vehicles.

Keywords: SUMO, Carbon Emissions, Fuel-Based Vehicles, Data-Driven Approach, Machine Learning

Chapter 1

Introduction

Chapter 1: Introduction

In this paper, we are working from the perspective of the Jashore city of Bangladesh, where these battery-powered vehicles have the most dominant share of the public transportation system. Although these battery-powered three-wheelers became the most popular means of transport, they have a big drawback of their uptime. As these vehicles are powered by batteries, they give a coverage of 60-100 km per charge. So, we tried to work on how battery-powered vehicles contribute much to a cleaner, greener city. For this work we modeled these battery-powered vehicles within a simulation environment using SUMO (Simulation of Urban Mobility). We have made a dataset by collecting real-world data on vehicle configurations like vehicle type, vclass, emission class, sigma, length, width, height, mass, acceleration, deceleration, max speed, min gap, departure time, and travel time. The emission output dataset has important parameters like vehicle CO₂, vehicle CO, vehicle HC, vehicle NO_x, and vehicle PM_x. Then we used machine learning algorithms to optimize the emission of these vehicles [4]. The model will analyze emission rates and how electric vehicles impact the environment. This work will be a great tool for extending the range of battery-powered vehicles. It will also reduce air pollution and give more freshness. So, it will greatly help the battery-powered vehicle sectors in the urban areas of Bangladesh.

1.1 Background

Urban population growth in developing countries has led to a rapid rise in transportation demand, especially in city centers. There are many fuel-based vehicles that emit carbon continuously in the environment [5]. Many cities struggle to provide affordable and sustainable public transportation for their expanding populations. In response to these challenges, environmentally friendly and cost-efficient alternatives to traditional fossil-fuel vehicles are being sought worldwide. One promising development is the adoption of electric battery-powered vehicles for public transport, which produce no tailpipe emissions and can offer a cheaper mode of travel for the masses.

In Bangladesh and similar contexts, low-cost battery-powered three-wheeler vehicles (such

as easy bikes and electric rickshaws) have become an integral part of urban transportation. These vehicles are popular due to their accessibility, low operating cost, and ease of use, effectively filling gaps in the public transit system. However, a significant drawback of these electric vehicles is their limited driving range or uptime on a single battery charge—typically around 60–100 km. This constraint leads to frequent downtime for charging, reducing the vehicles’ availability and earning potential for drivers. The limited range, combined with a lack of strategic route planning, means that energy is often used inefficiently during daily operations. There is a clear need for optimizing the battery-powered vehicles to minimize carbon emissions. By finding the impacts of carbon, air pollution can be reduced, effectively extending the vehicles’ usable range and improving their overall service uptime.

1.1.1 Definition

1.1.1.1 Battery-Powered Three-Wheeler

A small, motorized vehicle with three wheels that runs on an electric battery instead of a combustion engine. This category includes vehicles like easy bikes and electric rickshaws, which are used as low-cost public transport in many Asian cities. These vehicles produce no direct emissions and typically carry several passengers at a time.

1.1.1.2 Easy Bike

A locally used term in Bangladesh for a type of electric three-wheeler taxi. Easy bikes are battery-powered vehicles, usually seating around 4–6 passengers, and are widely used for short-distance transportation in urban areas. They have relatively low top speeds and are valued for being affordable and environmentally friendly.

1.1.1.3 Electric Rickshaw

An electric three-wheeled rickshaw, similar in purpose to an e-bike but often resembling the traditional pedal-powered rickshaw in design. It is powered by a rechargeable battery and can

carry a driver and a few passengers. Electric rickshaws provide a motorized alternative to cycle rickshaws, reducing human effort and increasing speed while still being emission-free at the point of use.

1.1.1.4 Carbon Emissions

In the context of battery-powered vehicles, "carbon emission" refers to no emissions from these electric three-wheeled vehicles. They emit no type of greenhouse gases (GHGs) like carbon monoxide, carbon dioxide, hydrocarbons, nitrogen oxides, particulate matter, and so on [6]. For this reason, battery-powered vehicles are more efficient for the environment and the people's health. This will ensure a greener country.

1.1.1.5 Route Planning

We are selecting many travel routes in Jashore city, and we designed these routes in SUMO [7]. We will detect carbon emissions from gasoline-based vehicles on the city road. For electric vehicles, they will also run on the same road as an alternative transportation. Then we will compare the importance of electric vehicles for both the people and the environment. The goal is to identify the impacts of the environment after adopting the battery-powered electric vehicles on the city roads.

1.1.1.6 Simulation of Urban Mobility (SUMO)

An open-source traffic simulation software package that allows researchers to model and simulate road networks, vehicle movement, and traffic behavior. SUMO provides a virtual environment to test traffic scenarios, route planning strategies, and vehicle performance without real-world trials [7]. In this project, SUMO is used to create a model of Jashore's urban road network and to simulate the behavior of battery-powered vehicles under different routing strategies with the traditional gasoline-based vehicles.

1.1.1.7 Traffic Control Interface (TraCI)

A software interface that enables external programs to interact with a SUMO simulation in real time. TraCI (often used with Python) allows for retrieving simulation data (such as a vehicle's speed, position, or battery charge at each timestep) and for dynamically controlling aspects of the simulation (such as vehicle routing or traffic signals) [7]. It was used in this study to extract vehicle telemetry data during simulations for analysis and to implement dynamic interactions with the simulated vehicles.

1.1.1.8 Machine Learning

A branch of artificial intelligence involving algorithms that learn patterns from data and make predictions or decisions. In the context of this thesis, machine learning techniques are employed to analyze complex relationships between route characteristics and carbon emissions. By learning from simulation results (and potentially real-world data), the model can predict which vehicle types are most impactful for carbon emissions.

1.1.1.9 Induction Loop Detector (E1 Detector)

An E1 detector simulates a physical induction loop placed at a fixed point on a lane. Whenever a vehicle passes over the loop, the detector records it. It captures instantaneous, point-based data, meaning it only detects vehicles exactly when they cross the loop location. It measures vehicle count, vehicle speed, vehicle length, time gaps, and occupancy (percentage of time the loop is covered). It helps generate traffic counts like real-world loop detectors, useful for calibrating traffic flow [7]. It is also often used in evaluation, traffic control, or adaptive signal control setups and works very well for point-based monitoring.

1.1.1.10 Lane Area Detector (E2 Detector)

An E2 detector simulates an extended detection zone along a lane or part of a lane (rectangle or polygon). It detects vehicles that enter, leave, or remain inside the detection area. It provides

area-based monitoring, not just point readings [7]. It measures the number of vehicles currently in the area and vehicle occupancy (how much of the detector area is covered). It is used in traffic signal control and vehicle stop-and-go behavior.

1.1.1.11 TLS (Traffic Light System)

A TLS is SUMO's internal term for traffic light controllers that manage traffic lights at intersections [7]. TLS represents a junction with one or more traffic signals, a program of signal phases (green, yellow, and red), and the timing and order of phases.

1.2 Problem Statement

The central problem addressed in this research is that fuel-based vehicles emit lots of carbon in the environment day by day. As for this, global warming has occurred, and the temperature of air is also increasing. Global warming refers to the long-term increase in Earth's average temperature due to the buildup of greenhouse gases (GHGs) in the atmosphere, mainly carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O). Carbon emissions affect the whole world, like rising global temperatures, melting glaciers and polar ice, sea-level rise, extreme weather events, agricultural impacts, health impacts, loss of biodiversity, etc. The challenge, therefore, is to develop a plan for battery-powered electric vehicle adoption among the people of our country that minimizes the carbon emissions so that it will be a safer place to live for both humans and animals in nature.

1.3 Objectives

- To develop a realistic simulation model of an urban transport network (using SUMO) that incorporates low-cost battery-powered vehicles and their operational parameters.
- To identify and analyze the factors of fuel-based vehicles that emit carbon and differentiate them from the battery-powered three-wheelers on various routes.

- To design an optimization algorithm (using machine learning and/or heuristic methods) that can determine the different amounts of carbon that are emitted from fuel-based vehicles.
- To implement an induction loop detector (E1 detector) and lane area detector (E2 detector) that help to generate traffic counts and measure the number of vehicles currently in the area for real-time telemetry monitoring.
- By adopting these electric vehicles by the people, we will find out the effect on the environment in our country.

1.4 Significance of the Study

The outcomes of this study have practical significance for sustainable urban transportation in developing regions. By extending the range of battery-powered public transport vehicles, the proposed approach that shows how carbon emissions affect the environment and how these electric vehicles can be adapted by the people of our country can directly benefit both the people and animals. In the context of Bangladesh's cities (such as Jashore), where electric three-wheelers are a major mode of transport, these improvements can enhance the overall efficiency and affordability of the urban transit system.

Beyond the immediate local benefits, the study contributes to broader goals of carbon emissions and environmental sustainability. This research also demonstrates how data-driven strategies (combining traffic simulation, real-world data, and machine learning) can address transportation infrastructure challenges. The integration of adapting electric vehicle communication in the proposed solution provides a forward-looking example of smart city technology improving transportation. In academic terms, the work fills a knowledge gap by focusing on adoption for informal battery-powered electric vehicles, offering insights that can be applied to similar transportation challenges in other cities and laying groundwork for future research in intelligent transport systems.

Chapter 2

Literature Review

Chapter 2: Literature Review

The field of electric vehicle (EV) adoption has evolved rapidly, moving from foundational algorithmic work focused on carbon dioxide minimization to complex, multi-objective systems that integrate real-time data and urban infrastructure. And the lower price of these EVs can make it a huge scope to establish this as an alternative transportation sector [8]. This review synthesizes the contributions of recent work, organized chronologically by the complexity of the models and objectives addressed.

2.1 Data-Driven Advancements of AI and Adoption Sector

Machine learning (ML) has become an essential tool for analyzing, predicting, and optimizing carbon emissions across various sectors, including transportation, energy, industry, and urban systems. In my thesis paper, I would consider carbon emissions from the transportation sector. Existing literature highlights three major applications: emission prediction, emission monitoring, and emission reduction optimization.

This study introduces a deep learning approach for forecasting CO₂ emissions from traffic vehicles using deep learning models like long short-term memory (LSTM) and bidirectional LSTM (BiLSTM). They introduced a vehicular emission dataset, which has many attributes like engine size, fuel type, consumption patterns, etc. For handling the outliers, they used the k-means clustering technique. The BiLSTM model performed better than the LSTM model and achieved a high R² score of 93.78%, which indicates excellent prediction accuracy. There are some metrics, such as MSE, RMSE, and Pearson's R, that were used to verify the model's performance. This study's main aim is to support planning smart cities and environmental policymaking in high-carbon-emission regions [9].

This research aims to reduce emissions and develop sustainable transport. They focus on the transportation sector, which is more responsible for CO₂ emissions. This study applies three deep learning algorithms, like Multi-layer Perceptron (MLP), Extreme Gradient Boosting (XGBoost), and Support Vector Machine (SVM), to predict the CO₂ emissions. The statistical indicators' R² values range from 0.8969 to 0.9886, and RMSE values range from 0.0333 to

0.1007. High fossil fuel consumption is so responsible for growing pollutants in the atmosphere, which worsens the air quality on a global scale. Due to limited resources of energy, the energy demand will be more efficient for future energy resources. This study is restricted to Turkey, which influences the CO₂ emissions in the transportation sector [10].

This paper discusses how machine learning and AI can evaluate CO₂ emissions from various sectors. It mainly focuses on the transportation sector for emissions, using many vehicle attributes and driving characteristics. Machine learning models like SARIMA (SARIMAX), based on ARIMA, are used in this research. The models emphasize the result and give an idea of effectiveness in the environment. The environmental impacts are generalized by AI for the results. For the carbon footprint and emission forecasting monitoring, the author suggests a broader application [11].

This paper focuses on the environmental effect after the adoption of electric vehicles (EVs) by analyzing the life cycle of CO₂ emissions with machine learning algorithms. The author makes models for estimating emissions through various stages like manufacturing, usage, and disposal. Though EVs are clean, the upstream emissions must be considered. In this study, they used machine learning algorithms like Linear Regression (LR), Random Forest (RF), Support Vector Regression (SVR), and Bi-LSTM for prediction and accuracy. These models show a better result and effectively predict the life cycle of carbon emissions. But the Bi-LSTM model performs well among the other models. These findings give importance to indirect emissions in sustainable planning [12].

This paper proposes a deep learning method for predicting vehicular carbon emissions. For accurate prediction, this study maintains environmental data, types of vehicles, and travel patterns. The On-Board Diagnostics-II (OBD-II) dataset is used to access emission trends using a deep neural network model. A 2-layer Long Short-Term Memory (LSTM) model is used in the proposed model. It gives better results and accuracy than traditional machine learning models like linear regression, random forest, gradient boosting, support vector machines, and so on. This model can help in the urban areas with reducing transport-related emissions. This study

supports the policy of making AI-driven environmental monitoring [13].

In this study, the author rates vehicles based on their CO₂ emissions by using supervised machine learning algorithms. Fuel consumption, engine size, and other specifications—they classify the emission levels using these data. Supervised machine learning algorithms are tested for the result and compared. There are about 203 g/km of CO₂ emitted from the new passenger cars. The average CO₂ emissions were about 170 g/km. This research gives a solving direction of aware vehicle practices on roads that should be guided by the consumers and regulators. These models give high accuracy and are suitable for real-time applications. In urban traffic control systems, it also suggests integrating these models [14].

In this paper, the author applies many regression-based machine learning models (linear regression, Lasso regression, and ridge regression) for predicting the CO₂ emissions from the vehicles. For training, they used the “CO₂ Emissions_Canada.csv” dataset that contains many vehicle specifications. For each algorithm, they calculate the accuracy, R-squared, mean absolute percentage error (MAPE), and root mean square error (RMSE) for the comparison. They find an optimal balance between accuracy and overfitting by using ridge regression. For data-driven policy recommendations in the automobile sector, this research is more helpful [15].

This paper represents a machine learning-based approach to predicting vehicular carbon emissions using many features like vehicle type, engine specifications, fuel type, weight, driving patterns, environmental conditions, etc. They tested many machine learning algorithms, such as decision trees, random forests, linear regression, neural networks, and support vector machines. Among these random forest algorithms, the most accurate predictions are given. This study’s focus is to improve environmental impact monitoring through these methods. By integrating these models into automobile industries and environmental agencies, it will be greatly beneficial for all of us. These models also support the vehicular emission certification processes [16].

This paper discusses futuristic cities like NEOM (Saudi Arabia’s smart city project) to predict the carbon emissions using machine learning algorithms. They compare many machine learning algorithms like LogitBoost ensembles and decision trees and explore urban parameters

for forecasting CO₂ emissions. They analyzed the “Fuel Consumption” dataset from Kaggle. The best accuracy comes from the decision tree technique at 99.7%. About 99.3% accuracy was also achieved by the LogitBoost Ensemble technique. The goal is to help by planning a sustainable infrastructure in the urban areas. This study promotes a solution for energy efficiency and forecasting carbon dioxide in urban areas [17].

In this article, the author’s major concern is gradually released greenhouse gas (GHG) emissions. The main victim of global warming is CO₂. They explore a deep learning hybrid model of a convolutional neural network and a long short-term memory network (CNN-LSTM) to predict the CO₂ emissions, which focuses on India’s transportation data. They integrate historical emission trends and weather data for enhanced prediction of carbon emissions. This model gives better accuracy over traditional machine learning models and single deep learning models. This study shows India’s growing need for emission control in the urban and rural areas. This study suggests policy-level emissions forecasting by using hybrid AI systems and is also applicable for smart city planning [18].

In this paper, their main focus is examining the barriers and enabling factors that are highly related to electric vehicle (EV) adoption. In developing countries, they review many EV projects and policies. This paper analyzes comparative policies, assessments of infrastructure, and developing a five-factor dependent framework for EV adoption. In this study, they found five major pillars for EV adoption. These pillars are policy, infrastructure, economic incentives, technology readiness, and institutional support. They also found out the barriers to EVs in developing countries. These are weak grid infrastructure, the high purchasing cost of EVs, limited charging stations, and governmental policy inconsistency [19].

This paper analyzes the economic and psychological factors that influence electric vehicle (EV) adoption. For this, they consider consumer attitudes (mediator) and demographic (moderator) factors. They surveyed a total of 288 EV users to find out their economic conditions. They proposed the Structural Equation Modeling (SEM) technique for the mediation and moderation analysis. For EV adoption, consumer attitudes significantly intermediate economic factors. De-

mographics (age, income, education) are the significant factors for adoption. They analyzed that young and educated consumers are willing to adopt the EVs. On the other hand, older and low-income consumers are not willing to adopt the EVs due to their financial condition [20].

They investigate EV deployment in China and how it contributes to China meeting its carbon neutrality goal by 2060. For this reason, they established transport energy models, bottom-up technology models, and transport demand modeling. They analyzed many scenarios and economic cost-effectiveness for EV adoption. By adopting EV, the market share has significantly increased. It reduced the higher emissions. Northeastern and northwestern provinces have lower cost-effectiveness and lower emission reduction benefits [21].

This paper develops a network-based approach of adoption to understand and control mass EV adoption. To analyze EV adoption, they proposed some methods like data-driven analysis of mobility habits, social network modeling, proximity-based network construction, cascade diffusion modeling, spread maximization framework, and policy cost-benefit simulation. To accelerate the adoption rate, influential nodes are a major concern. It provides a control-oriented framework for policymakers to adopt EVs [22].

In this paper, they predict the regional EV adoption using machine learning algorithms and socio-economic factors in the United States. They take the 2017 National Household Travel Survey (NHTS) dataset. For handling the dataset, they take steps. They used some machine learning algorithms like decision trees (DT) and other comparative ML models. From these models, decision trees (DT) performed well compared to other ML models. Higher gasoline prices are a barrier to EV adoption. Car sharing availability is very significant for adoption. It generated a regional EV penetration map, like the United States [23].

Chapter 3

Methodology

Chapter 3: Methodology

3.1 Research Design

The research follows a sequential, modular design that integrates simulation, real-time data logging, and data analysis. Figure 3.1 below illustrates the step-by-step approach used to develop and evaluate carbon emissions from fuel-based vehicles in Jashore City.

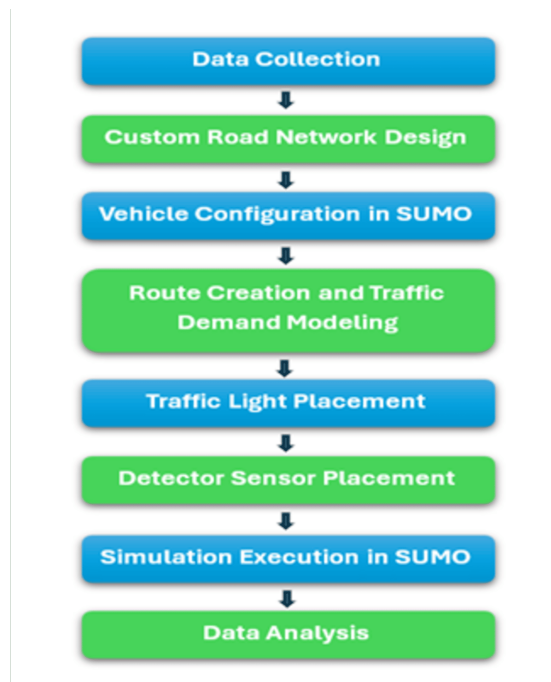


Figure 3.1: Research Design Flow

3.1.1 Data Collection

The process begins with collecting real-world specifications of ten different types of fuel-based vehicles (passenger cars, buses, trucks, taxis, emergency vehicles, bikes, delivery vehicles, coaches, SUVs, and trailers). Data was gathered from manufacturers and verified with local operators, covering vehicle dimensions and engine behavior.

3.1.2 Custom Road Network Design

A custom road network was constructed in Figure 3.2 using OpenStreetMap data and refined in NetEdit to capture six critical junctions in Jashore city [24]. The road geometry, junction curvature, and directional flow were tailored for simulation fidelity [7].

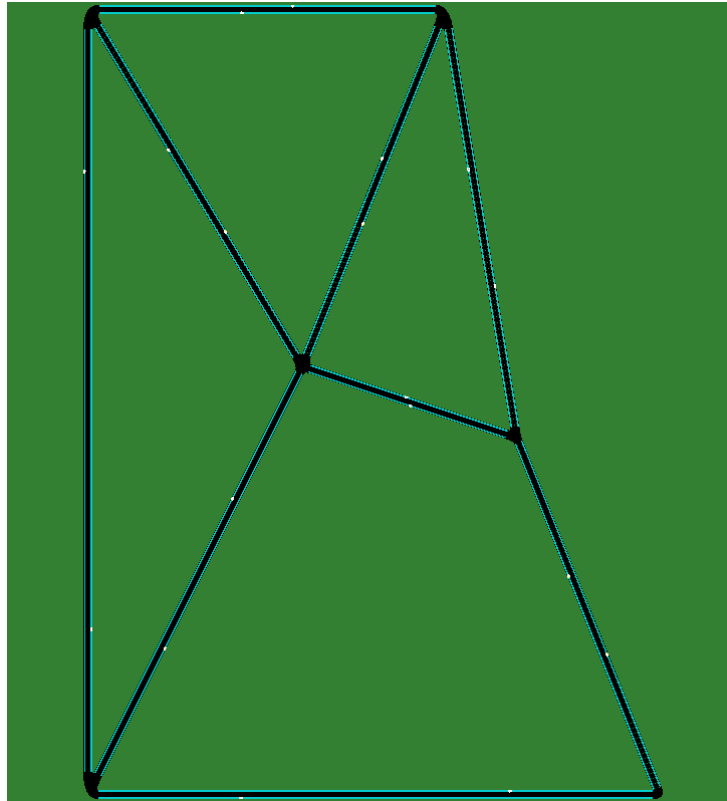


Figure 3.2: Custom Road Network Design

3.1.3 Vehicle Configuration in SUMO

Each vehicle model was configured within SUMO using realistic parameters such as vehicle's type, class, acceleration, deceleration, mass, sigma, length, width, height, maximum speed, minimum gap, emission class, lcStrategic, lcCooperative, lcSpeedGain, lcKeepRight, and laneChangeMode [7]. These configurations ensured accurate carbon emissions during route simulation. Table 3.1 shows the configuration of vehicles.

Table 3.1: Vehicle Type Parameters Used in Simulation

Parameters	Passenger	Bus	Truck	Taxi	Emergency	Bike	Delivery	Coach	SUV	Trailer
vType	passenger	bus	truck	taxi	emergency	bike	delivery	coach	suv	trailer
accel (m/s ²)	2.6	1.2	1.3	2.2	2.5	5.0	2.5	1.5	2.5	1.2
decel (m/s ²)	4.5	4.0	4.0	4.0	4.2	10.0	4.2	4.2	4.2	4.0
mass (kg)	1500	10000	12000	1500	5000	200	5000	12000	5000	14000
sigma (s)	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
length (m)	4.5	10.0	7.0	4.5	5.0	2.0	5.0	12.0	5.0	7.0
width (m)	2.0	2.5	2.2	2.0	2.0	0.8	2.0	2.5	2.0	2.0
height (m)	1.5	3.0	2.5	1.5	2.0	1.2	2.0	3.0	2.0	3.0
minGap (m)	2.0	2.5	2.0	2.0	2.0	1.5	2.0	2.0	2.0	2.0
maxSpeed (m/s)	22.0	20.0	15.0	22.0	25.0	20.0	15.0	20.0	20.0	15.0
emissionClass	HBEFA4	HBEFA4	HBEFA4	HBEFA4	HBEFA4	HBEFA4	HBEFA4	HBEFA4	HBEFA4	HBEFA4
lcStrategic	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
lcCooperative	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
lcSpeedGain	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
lcKeepRight	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
laneChangeMode	1621	1621	1621	1621	1621	1621	1621	1621	1621	1621

3.1.4 Route Creation and Traffic Demand Modeling

Nine distinct routes were designed to reflect common urban travel patterns, varying in distance and complexity. Vehicle trips were synchronized in simulation to enable comparative analysis of carbon emissions across different routes.

3.1.5 Traffic Light Placement

Five strategically placed traffic lights in Table 3.3 were modeled in the simulation to reflect actual traffic lights on Jashore Road [7]. Each traffic light was configured with specific characteristics like stop, hold, run, and vehicle delays.

3.1.6 Detector Sensor Placement

There were two types of detector sensors, such as the E1 Detector (Induction Loop Detector) and E2 Detector (Lane Area Detector), integrated in each lane that collect vehicle types that run in the routes [7]. In table 3.5 it is shown partially.

3.1.7 Simulation Execution in SUMO

The entire simulation, including vehicle movement, speed, behavior, and detector sensor, was executed in SUMO 3.4. Simulation step size was set to 1.0 seconds for high-resolution tracking of battery consumption.

3.1.8 Data Analysis

Post-simulation data was exported using custom Python scripts and the CSV format. Emissions, car information, a summary, and vehicle counts were generated. These insights serve as the foundation for efficiently adapting vehicle path planning algorithms.

3.2 Data Collection

In Jashore three-wheeler battery-powered vehicles are the most common form of public transportation. The number of autorickshaws is increasing day by day. In figure 3.3, we can see how EV numbers are rapidly growing. Among them, we have selected a total of ten different types of combustion engine fuel types of vehicles, including passengers, buses, trucks, taxis, emergency vehicles, bikes, delivery vehicles, coaches, SUVs, and trailers. These fuel-based vehicles are commonly operated in Jashore City. So, we have collected their detailed specifications from both manufacturers and local operators and owners.

3.3 Implementation

3.3.1 Environment Setup

For this work, we used Eclipse SUMO (Simulation of Urban Mobility) for the simulation environment. For designing the network, we have used NetEdit, and for real-time data extraction, TraCI (Traffic Control Interface) is used. This simulation environment was set up in a Windows

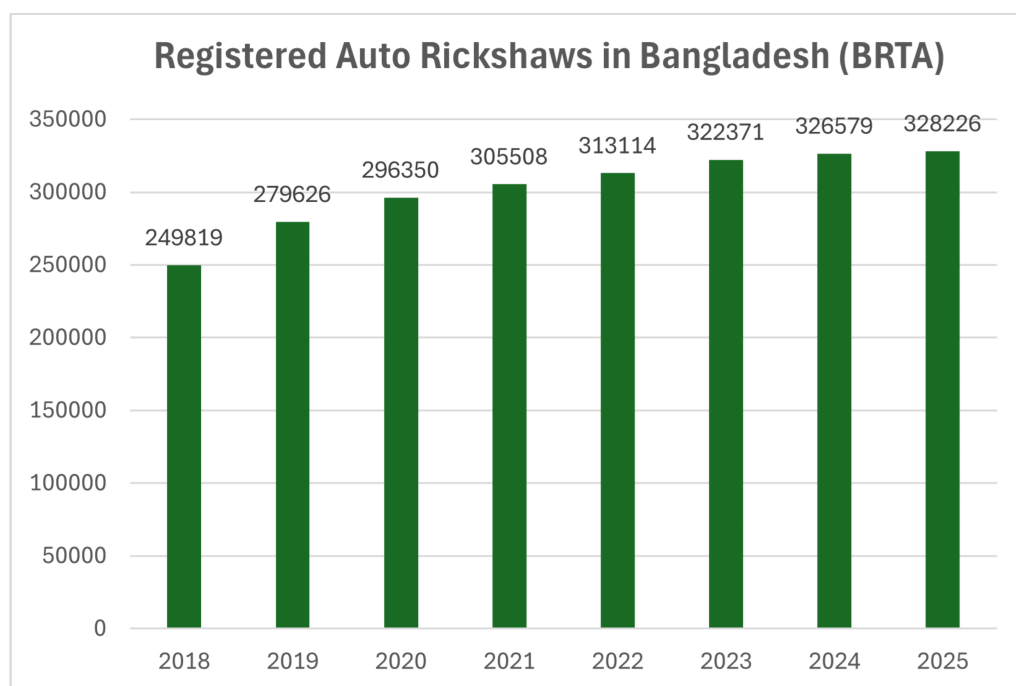


Figure 3.3: Registered Auto Rickshaws in Bangladesh

11 operating system with the environment variable SUMO_HOME set to enabled. For exporting real-time data from TraCI to an Excel file, we have used the pandas library. Here all the configuration files are written in Python 3.

3.3.2 Custom Road Network Creation

We have created a custom road network in Figure 3.2 of some important points of the Jashore city of Bangladesh. Jashore is a busy city, as it is an important transport corridor of Bangladesh. We have used the OpenStreetMap (OSM) figure 3.3 to follow the base road layout [24]. We have designed the road network in NetEdit to keep the necessary roads [7]. We have tried to match road geometry with the real world of the roads. Our custom road network includes 6 of the most important junctions of Jashore city: Palbari, Chachra, Doratana, New Market, Monihar, and Muroli Mor. These junctions are connected through 18 directional road edges.

3.3.3 Vehicle Configuration in SUMO

We have configured the vehicles with our collected parameters in the sumo configuration file. We have added general attributes like vehicle type, vClass, acceleration, deceleration, mass, sigma (driver imperfection), length, width, height, maximum speed, minimum gap, and emission class. For the overtake of the vehicles, there are some parameters like lcStrategic, lcCooperative, lcSpeedGain, lcKeepRight, and laneChangeMode [7]. We take the value of 1.0 for lcStrategic, lcCooperative, and lcSpeedGain. The value of lcKeepRight ranges from 0.3 to 0.6. For this simulation, we take this value to be about 0.3. The last parameter is lcSpeedGain. It helps the vehicles to overtake smoothly. We initially take this value as 1621 for each type of vehicle configuration. Table 3.1 shows the parameters that have been used for configuring vehicles.

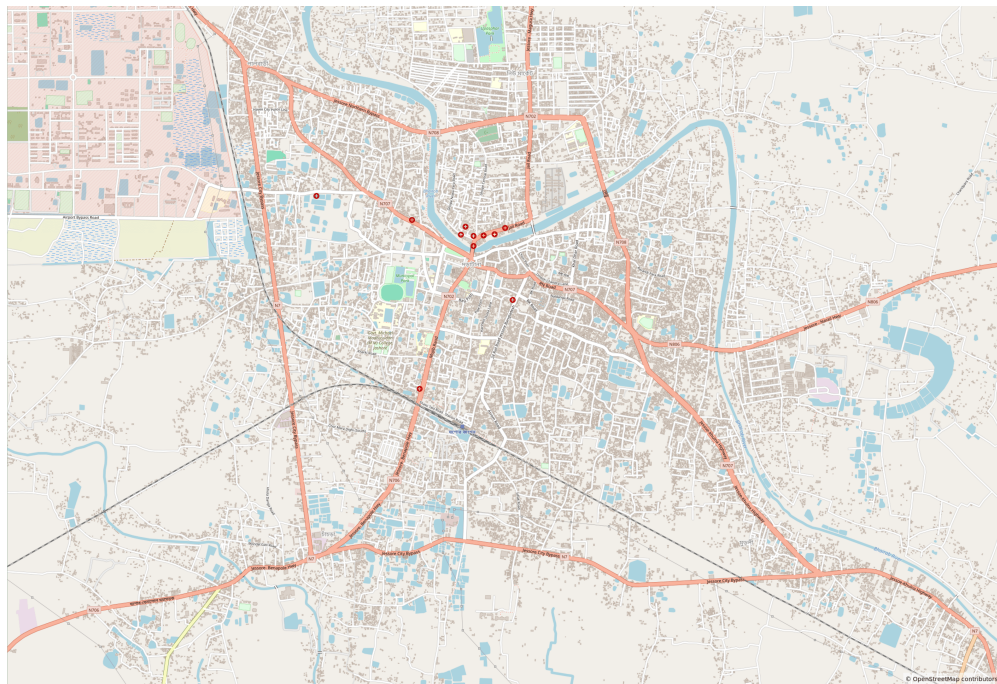


Figure 3.4: Map of Jashore City

3.3.4 Route Creation and Traffic Demand Modeling

We have configured a total of one hundred unique vehicle routes in figure 3.2 in the custom road network. Each of these routes represents a distinct directional flow. We have designed the routes in such a way that they cover major inter-junctional travel paths. We have defined the route considering the real-time traffic patterns. Here we mainly cover three types of routes: routes representing short point-to-point trips, routes covering medium-distance travel with junction transitions, and routes covering longer trips across multiple junctions and road segments. We have configured each route pattern to ten different types of fuel-based vehicles like passengers, buses, trucks, taxis, emergency vehicles, bikes, delivery vehicles, SUVs, trailers, and coaches, resulting in hundreds of individual vehicle entries in the route file.

The connected edges are shown in table 3.2. In this configuration all the vehicles depart at simulation time 0.00s, which ensures simultaneous operation for comparative analysis. The route file includes departure times, vehicle type parameters, and route IDs. The total travel time for all types of vehicles to finish their route is about 2853 seconds. The most critical finding is the successful measurement of carbon emissions, which is a direct result of configuring vehicles accurately according to the parameters in SUMO. The collected dataset enables the calculation of crucial statistics of vehicle CO₂, which are essential inputs for the optimization of machine learning algorithms.

3.3.5 Traffic Light Placement

We have placed traffic lights at the major five junctions on Jessore Road. In our custom road network design of Jashore, we have placed a total of five traffic lights in table 3.3 in a strategic way along the major corridors to support the operation of different types of vehicles. Here we have placed these traffic lights at important junctions, including Palbari, Doratana, New Market, Chachra, Muroli Mor, and Monihar, covering both directions. The vehicles will stop when the red lights are on and continue their routes without any disturbance. We placed an induction

Table 3.2: Route Connections and Distances

Route ID	Connected Edges	Distance (km)
A01	Palbari → New Market	2.55
B01	New Market → Palbari	2.55
A02	New Market → Monihar	2.30
B02	Monihar → New Market	2.30
A03	Monihar → Muroli Mor	2.60
B03	Muroli Mor → Monihar	2.60
A04	Muroli Mor → Chachra	4.30
B04	Chachra → Muroli Mor	4.30
A05	Chachra → Palbari	4.10
B05	Palbari → Chachra	4.10
A06	Palbari → Doratana	2.70
B06	Doratana → Palbari	2.70
A07	New Market → Doratana	1.60
B07	Doratana → New Market	1.60
A08	Monihar → Doratana	1.60
B08	Doratana → Monihar	1.60
A09	Chachra → Doratana	2.90
B09	Doratana → Chachra	2.90

Table 3.3: Traffic Light Placement

Node	Junction
0	Palbari
1	New Market
2	Monihar
4	Chachra
5	Doratana

loop detector (E1 detector) and a lane area detector (E2 detector) for the vehicles overtaking smoothly by the side lane.

3.3.6 Simulation Execution in SUMO

The simulation was executed using sumo-gui 3.4 for visual inspection and sumo for performance runs. The configuration file here combines the custom road network (file.net.xml), traffic demand routes (file.rou.xml), detector sensors (detectors.add.xml), and traffic light systems (tls.add.xml). Simulation step length was set to 1.0 seconds. We also defined delays to ensure

proper initialization. A total of ten types of fuel-based vehicles were simulated in the custom road network according to predefined route patterns.

3.3.7 Real-Time Data Extraction Using SUMO

The TraCI API was used to extract vehicle data during the simulation. For this, we have initialized Python connected to SUMO using the `sumo.cfg` configuration file. After starting the simulation, we collected vehicles at each simulation step. We have logged the data in an Excel file after the simulation ends, as shown in Table 3.4. In Table 3.4, we put the partial information on emissions from ten types of fuel-based vehicles with their total travel time. The simulation run in SUMO is shown in figure 3.5.

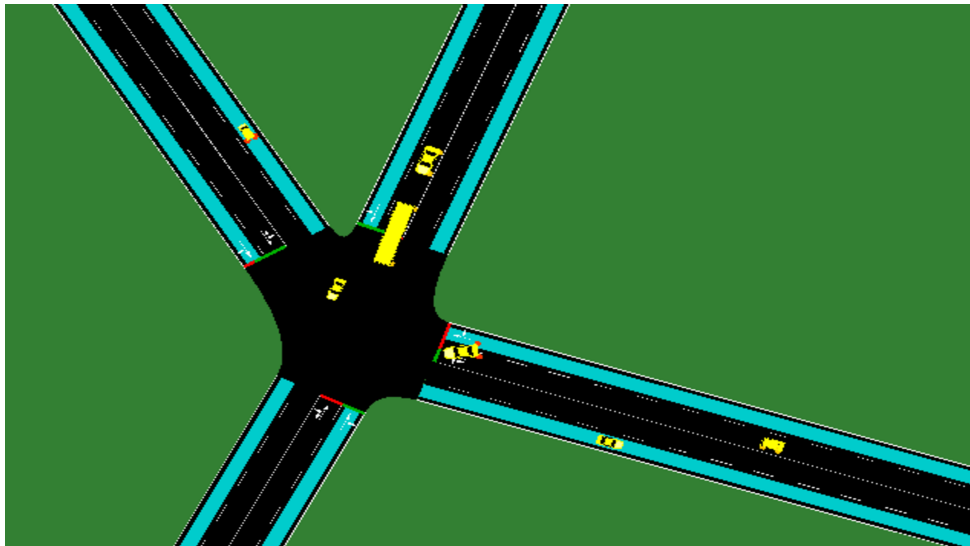


Figure 3.5: Simulation Execution in SUMO

3.3.8 Detector Sensor Implementation

For the vehicle counts of different types in the whole simulation, we placed an induction loop detector (E1 detector) and a lane area detector (E2 detector) that counts the vehicles by their types. There are a total of ten types of vehicles running in the SUMO simulation. Vehicle counts are shown in table 3.5.

Table 3.4: Data extraction from Emission File (Partial Example)

vehicle_id	vehicle_type	depart_time_s	travel_time_s	CO2_mg
0	passenger	1	214	432883.72
1	bus	2	101	186549.67
2	truck	3	169	319260.18
3	bus	4	104	217430.99
4	taxi	5	132	250753.27
5	trailer	6	258	505656.34
6	emergency	7	229	449420.69
7	taxi	8	170	336957.45
8	bus	9	2853	4404468.05
9	trailer	10	1374	2122456.68

Table 3.5: Vehicle Counts by Type

Vehicle Type	Count
Passenger	18
Bus	17
Truck	09
Taxi	10
Emergency	11
Bike	05
Delivery	11
Coach	08
SUV	06
Trailer	05

This network design achieved 100% vehicle coverage of the simulated routes. Data is collected at a 1-second interval during the simulation.

3.3.9 Machine Learning Algorithms Implementation

This research follows a sequential, modular design that integrates simulation, real-time data logging, and data analysis. The flowchart 3.1 below illustrates the step-by-step approach used to develop and evaluate carbon emissions from fuel-based vehicles in Jashore City.

After simulation, the dataset has been created. The next step is to develop some different machine learning models like linear regression, random forest regressor, gradient boosting, and XGBoost. We evaluate these models to predict the emissions produced by the different types

of fuel-based vehicles. In this paper, we consider these four models to compare accuracies, F1 scores, R^2 , MAE, MSE, RMSE values, and the capability to fit with the created dataset.

Linear Regression:

A linear regression model is a linear relationship between the independent variables and the target variables. In my dataset, vehicle features are independent variables, and CO₂-mg is my target variable. It assumes that:

$$CO_2 = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon \quad (3.1)$$

Where:

p = number of features,

x_j = independent variables,

β_j = regression coefficients

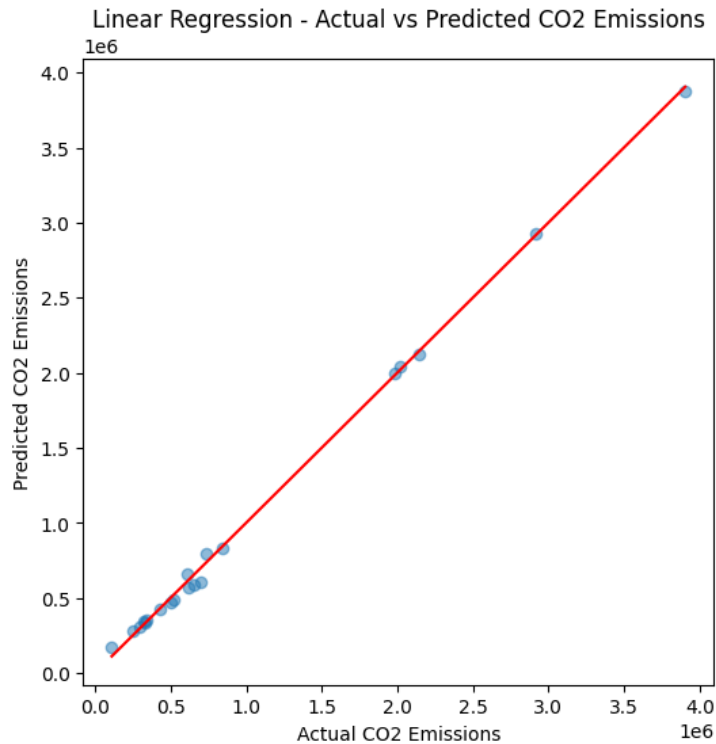


Figure 3.6: Linear Regression Model

This equation finds coefficients that minimize the mean squared error (MSE) values. The

CO₂ emission changes linearly with the vehicle features. It is fast and simple with linear relationships. It works well with this dataset. The accuracy of this model is about 92.72%. It gives better results. In this model the random state is 42 and the test size is 0.2, which is pipelined to maximize the output data. It is a good baseline model for prediction. About a 0.9984 R² value in this model. The actual vs. predicted CO₂ emissions graph for the linear regression model is shown in Figure 3.6.

Random Forest Regressor:

A random forest regressor is commonly used in supervised machine learning models that take the output from multiple decision trees. It averages the prediction values to get the final output. In this model, the random state is 42, max depth is 10, and total estimators are 200. This model gives better accuracy than the linear regression model. About 93.15% of accuracy comes from this model. In my dataset, this model captures the nonlinear relationships between vehicle parameters and CO₂ mg parameters.

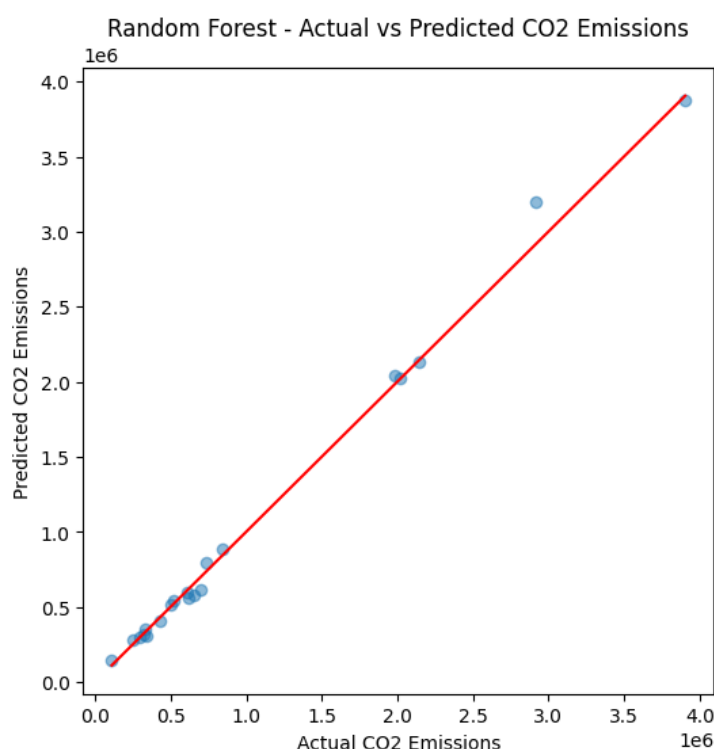


Figure 3.7: Random Forest Model

This model performs well with many features and handles complex interactions automatically. The R^2 value is about 0.9944 in this model. Random forest regressors usually perform well with emission datasets. The actual vs. predicted CO₂ emissions graph for the random forest model is shown in Figure 3.7.

Gradient Boosting (GB):

Gradient boosting algorithms sequentially build trees. The previous trees' errors are corrected by each new tree. It learns patterns gradually and optimizes loss function using gradient descent. This model is more sensitive to hyperparameters than the random forest model. In this model, the total number of estimators is 200, the learning rate is about 0.05, the max depth is 5, and the random state is 42. It gives much better accuracy. About 92.66% accuracy comes from this model. It produces much better accuracy and performance. The actual vs. predicted CO₂ emissions graph for the gradient boosting model is shown in Figure 3.8.

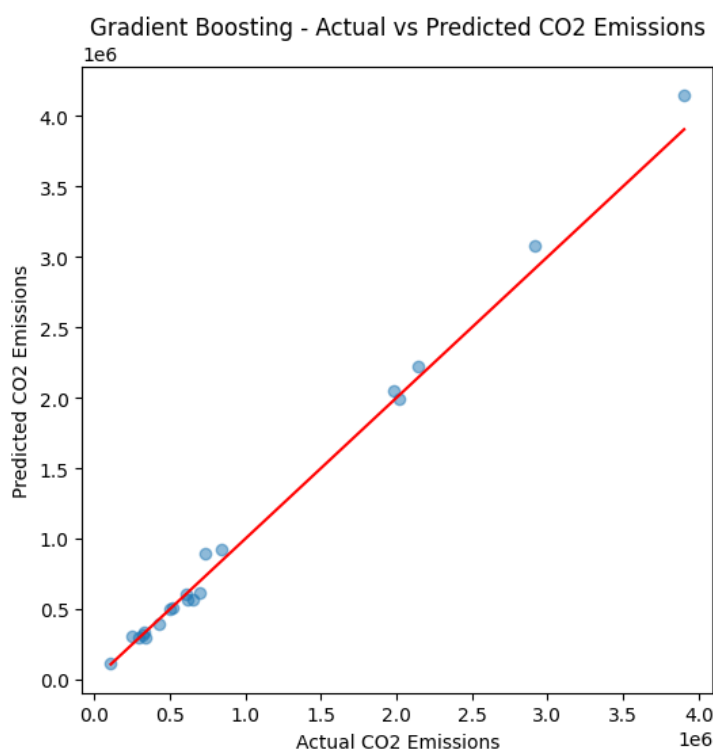


Figure 3.8: Gradient Boosting Model

XGBoost:

The XGBoost (Extreme Gradient Boosting) algorithm is an advanced version of the gradient boosting algorithm. For better performance we established this algorithm. It uses regularization, parallel processing, and true pruning and handles missing values automatically in this dataset. Almost 92.90% of accuracy comes from this model. This accuracy is the second highest accuracy value among these four models. RMSE values are higher in this model than in the other models. XGBoost is very efficient for large datasets and handles complex vehicular emission patterns. The actual vs. predicted CO₂ emissions graph for the XGBoost model is shown in Figure 3.9.

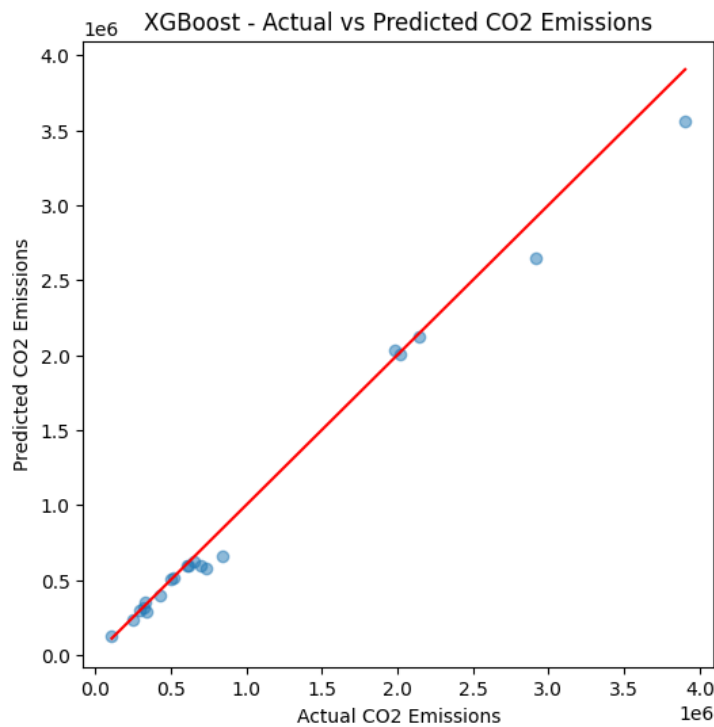


Figure 3.9: XGBoost Model

Data Collection and Analysis:

The simulation in SUMO outputs an emission file that emits different types of greenhouse gases (GHGs) from the fuel-based vehicles like passengers, buses, trucks, taxis, emergency vehicles, bikes, coaches, SUVs, trailers, and delivery vehicles. This emission file collects key telemetry: vehicle CO, vehicle CO₂, vehicle HC, vehicle NO_x, and vehicle PM_x. In the table 3.4, we extract partial data of the emissions from these vehicles.

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This chapter presents the quantitative results obtained from the simulation of the vehicle-to-infrastructure (V2I) data collection system. It provides a detailed analysis of these results, discusses their significance in the context of the research objectives, and explores the broader implications for electric vehicle (EV) adoption in Bangladesh. The presented findings reflect data collection, validating the core architecture and providing the necessary dataset for subsequent analysis and algorithm development.

4.1 Results

The simulation was executed to model the behavior of ten fuel-based vehicles over a 2853.00 second period within the defined road network of Jashore. The system successfully collected real-time telemetry data.

4.2 Discussion of Results

The results presented the vehicular emissions from the fuel-based vehicles compared to battery-powered electric vehicles. This thesis explores how electric vehicles (EVs) affect the environment and what benefits come from this. The discussion below interprets these findings in relation to the research goals.

This chapter presents the quantitative results obtained from the simulation of the vehicle-to-infrastructure (V2I) data collection system. It provides a detailed analysis of these results and discusses their significance in the context of the research objectives. The total travel time for all types of vehicles to finish their route is about 2853 seconds. The most critical finding is the successful measurement of carbon emissions, which is a direct result of configuring vehicles accurately according to the parameters in SUMO. The collected dataset enables the calculation of crucial statistics of vehicle CO₂, which are essential inputs for the optimization of machine learning algorithms.

We established four machine learning algorithms—linear regression, random forest regres-

Table 4.1: Comparison of Different Machine Learning Models

Model	R ²	MAE	MSE	RMSE	F1 Score
Linear Regression	0.9984	31643.81	1.620E9	40258.94	1.00
Random Forest	0.9944	45829.49	5.630E9	75023.27	1.00
Gradient Boosting	0.9923	61220.40	7.710E9	87843.07	1.00
XGBoost	0.9862	68865.49	1.38E10	117397.8	0.95

Linear regression, gradient boosting, and XGBoost. Among these models, we compare R², MAE, MSE, RMSE, and F1 Score. In the linear regression model, the R² value is higher than the other models. Both mean absolute error (MAE) and root mean squared error (RMSE) are high for the XGBoost algorithm. These values are compared in Table 4.1 for the most accurate predictions of these models. Table 4.1 is shown below:

The most accuracy comes from the random forest regressor model. About 93.15% accuracy in this model, which is higher than the other models. With the limitations and barriers, the random forest regressor model is the best fit for accurate prediction. The comparison of accuracy for each model is shown in Figure 4.1.

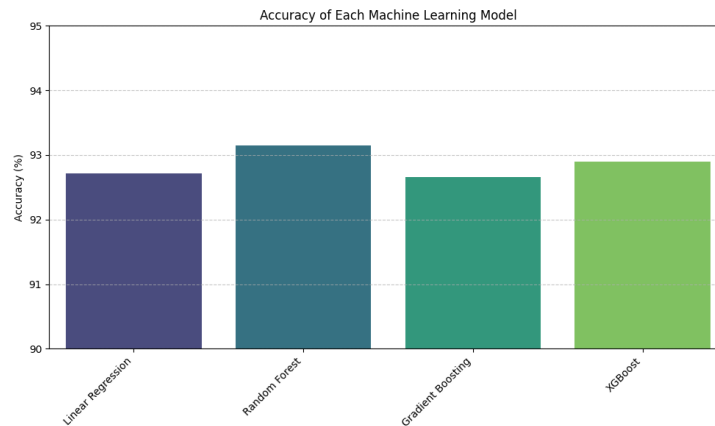


Figure 4.1: Accuracy Comparison Among Each ML Models

The heatmap of RMSE (Root Mean Square Error) comparison visually represents the prediction error magnitude. This comparison is among the machine learning models for CO₂ emissions estimation. The RMSE value is measured by the average deviation between the actual CO₂ emissions values and the model-predicted CO₂ emissions values:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4.1)$$

Where:

y_i = actual value,

$\hat{y}_i$ = predicted value,

n = total number of samples

A lower RMSE value gives better prediction accuracy. So, the linear regression model gives the better prediction accuracy as the value of RMSE is lower than the other models. The heatmap of RMSE comparison among the models is shown in Figure 4.2.

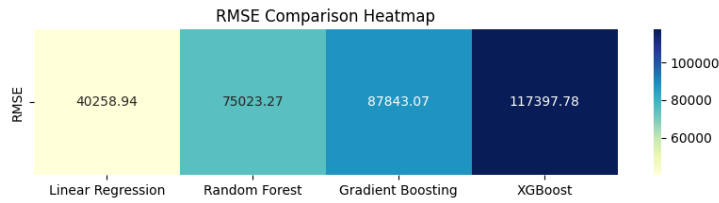


Figure 4.2: RMSE Comparison Heatmap

The correlation heatmap is another term that is very important. It shows the strength of linear relationships between the actual CO₂ emissions values and the model-predicted CO₂ emissions values. This correlation is based on the Pearson correlation coefficient:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad (4.2)$$

Where:

$\text{Cov}(X, Y)$ = covariance,

σ_X, σ_Y = standard deviations

The correlation heatmap between actual and predicted CO₂ emissions is shown in Figure 4.3.

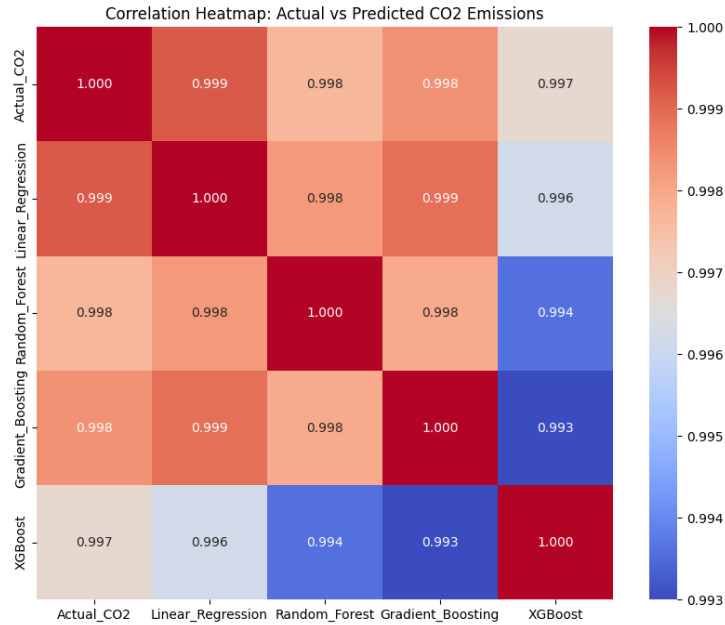


Figure 4.3: Correlation Heatmap Between Actual and Predicted CO₂ Emissions

4.2.1 Significance of Emission Data

The most critical finding is the successful measurement of carbon emissions, which is a direct result of configuring vehicles accurately according to parameters in SUMO. The collected dataset enables the calculation of crucial statistics like vehicle CO, vehicle CO₂, vehicle HC, vehicle NO_x, and vehicle PM_x, which are essential inputs for the optimization of many machine learning algorithms for the further development of my thesis.

4.3 Implications

The findings of this research have significant practical implications, providing a validated foundation for future work in smart mobility and EV adoption in Bangladesh.

4.3.1 Foundation for EV Adoption

The foundation of electric vehicle (EV) adoption is built on a combination of technological, economic, environmental, and policy-driven factors that encourage the transition from con-

ventional gasoline vehicles to cleaner electric mobility. At the core, advancements in battery technology, including improved energy density, reduced cost, and longer lifespan, have made EVs increasingly practical and affordable. Growing concerns over climate change and rising carbon emissions have strengthened public and governmental interest in EVs as a sustainable solution, supported by global commitments to reduce greenhouse gas emissions [25]. Economic incentives such as government subsidies, tax benefits, reduced registration fees, and lower operating costs further accelerate adoption. The expansion of charging infrastructure, including fast-charging networks and home-charging solutions, addresses range anxiety and enhances user convenience. Moreover, social factors—such as increasing environmental awareness, urban air pollution concerns, and positive consumer perceptions—play a crucial role in shaping demand.

4.3.2 Expansion of Charging Infrastructure

Our country is gradually expanding its electric vehicle (EV) charging infrastructure to support the country's growing interest in sustainable transportation. Although the current number of public charging stations is limited. Overpopulated cities like Jashore have begun installing the charging infrastructure to meet this demand. Despite challenges such as grid capacity constraints, high installation costs, and lack of standardization, Bangladesh's charging network is expanding steadily, reflecting the nation's commitment to promoting EV adoption and reducing long-term carbon emissions.

4.3.3 Smart Traffic Management System in Jashore

The validated V2I system is a foundational component of a larger intelligent transportation system (ITS). The real-time vehicle location and behavior data collected via TraCI and aggregated by the E1 and E2 detectors can be fed into a central traffic management center. This would allow authorities to monitor traffic flow and detect congestion in real-time, enabling potential interventions like dynamic signal adjustments.

4.3.4 Policy and Economic Implications

For a developing country like Bangladesh, this research proves that a robust V2I network can be built using open-source tools (SUMO, Python) and commodity hardware. This lowers the barrier to entry for government agencies and local startups to innovate in the smart mobility sector. The data can be used to formulate evidence-based policies, such as offering incentives for operating EVs or providing subsidies.

Chapter 5

Conclusion

Chapter 5: Conclusion

This chapter summarizes the key achievements of the research, outlines the limitations encountered during the implementation and simulation phase, and details the essential next steps required to complete the project, specifically focusing on the application of machine learning for optimizing carbon emissions and analyzing the EV adoption by the people of our country.

This research successfully designed, implemented, and validated a robust vehicle-to-infrastructure (V2I) data collection architecture utilizing the Eclipse SUMO simulation environment. The core objectives were achieved through the creation of a custom road network of Jashore City, Bangladesh; realistic configuration of ten different types of fuel-based vehicles (like passengers, buses, trucks, taxis, emergency vehicles, bikes, coaches, SUVs, trailers, and delivery vehicles); and the establishment of a functional data pipeline. The key successes of this study include:

- **V2I Architecture Validation:** The end-to-end system comprising SUMO simulation, TraCI data extraction, and detector units (E1 and E2) successfully collected a 2853-second record of GHG emissions from 100 vehicles over steps.
- **Complete Coverage and Redundancy:** The network design achieved 100% vehicle tracking coverage over the simulated routes, with each vehicle detected by E1 and E2 detector sensors, ensuring high data reliability.
- **Data-Driven Insights:** The detector sensor (E1 and E2) activity metrics clearly identified what type of vehicles run through custom roads, providing direct insights for future infrastructure and how electric vehicles will impact urban areas like Jashore.

In conclusion, this phase of the research validated the methodology for acquiring realistic data, laying a solid and verified foundation for the primary goal of electric vehicle adoption.

5.1 Limitations

The current phase of the research, being defense, is subject to the following limitations:

- **Simulation Scope:** The road network is limited to a custom creation of only 6 major junctions of Jashore City, connected by 18 road edges. This is a simplification of the real urban environment.

- **Static Traffic Demand:** The traffic demand model relies on only 18 unique predefined vehicle routes and configures all vehicles to depart simultaneously at 0.00s. This setup ensures comparative analysis but does not fully represent the dynamic, stochastic nature of real-world traffic flow.

- **Absence of Optimization/ Prediction:** The project is currently limited to data collection and validation. The core outcome of the research, the implementation and evaluation of emission data, and how much impact they have not yet been performed.

5.2 Recommendation and Future Scope

To fully realize the potential of the collected dataset and complete the research objectives, the following steps are critical:

5.2.1 Machine Learning Model Selection and Training:

We employed four machine learning (ML) models (linear regression, random forest, gradient boosting, and XGBoost) to train on the extracted data. The primary goal of these models is to perform accurately to predict the carbon emission from different types of fuel-based vehicles across different road segments and under various driving conditions (speed, acceleration, etc.). For more accurate predictions, there need to be established advanced machine learning models like LSTM. This study contributes a better understanding of the true environmental carbon emissions from fuel-based vehicles. It also promotes more sustainable decision-making in the transportation sector in our country.

5.2.2 Machine Learning Model Selection and Training:

The analyzing of carbon emissions has to be integrated into a broad area of our country. In this research, we consider only one city of Bangladesh named Jashore. In this paper, the road network is limited to the custom creation of only six major junctions of Jashore City, connected

by eighteen road edges. It configures all vehicles to depart simultaneously at 0.00s. This setup ensures comparative analysis but does not fully represent the dynamic, stochastic nature of real-world traffic flow. The goal is to identify the whole emission data that are emitted through different types of fuel-based vehicles in our country. Future research could be done by including broad real-world data of different regions in our country. It will expand the dataset and improve the machine learning models' performance. It will also be compared with the regional differences of carbon emissions from different fuel-based vehicles.

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